



Company Filings — Technical Documentation

Coresight Research Portal — Technical Documentation

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Application: Market Data Portal (MDP)

Page module: `app/pages/company_filings.py`

URL: `/company_filings`

Registered in: `app/main.py` as `( "pages/company_filings.py", "Company Filings", "company_filings", False)`

Product name: Company Filing Documents (Market Data Portal)

1. Purpose and scope

The Company Filings page is the primary SEC and non-SEC filing browser for the portal. It combines:

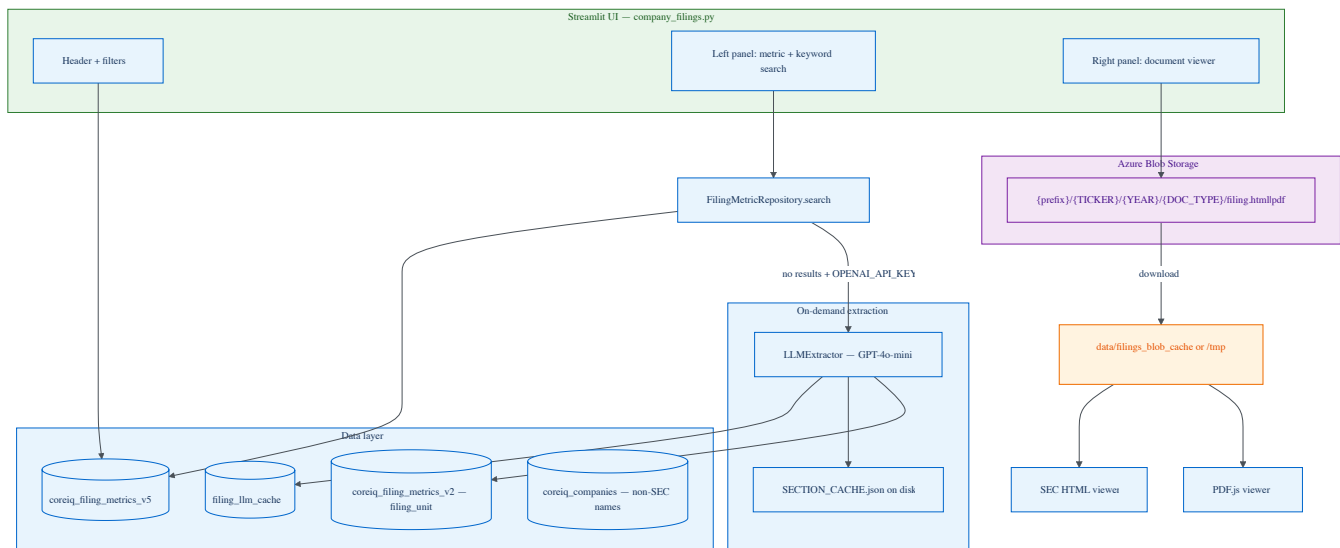
- **Filter-driven navigation** — company, document type, optional filing unit (8-K / 6-K / proxy), fiscal/storage year
- **Metric search** — pre-ingested eXtensible Business Reporting Language (XBRL) metrics from MySQL (`coreiq_filing_metrics_v5`)
- **Document viewing** — SEC ieXtensible Business Reporting Language (XBRL) HTML in a sandboxed iframe; non-SEC PDFs via PDF.js
- **Keyword search** — full-text hits in the loaded document (HTML or PDF)
- **LLM fallback** — GPT-4o-mini extraction when DB search returns no rows (`app/core/llm_extractor.py`)
- **Azure Blob Storage** — source of truth for filing HTML/PDF files

The page does **not** call `edgartools` at runtime. eXtensible Business Reporting Language (XBRL) data is ingested offline into `coreiq_filing_metrics_v5` ; metrics with `source='edgartools'` are displayed with an EDGAR badge when present in the DB.

2. High-level architecture

Filing viewer architecture: SEC eXtensible Business Reporting Language (XBRL) path, non-SEC blob PDFs, and LLM fallback extraction.

Filing viewer architecture: SEC eXtensible Business Reporting Language (XBRL) path, non-SEC blob PDFs, and LLM fallback extraction.



3. UI layout and controls

3.1 Page structure

Region	Width	Contents
Global header	Full width	<code>render_header(..., current_page="company_filings")</code>
Title row	1/3 + 2/3	"Company Filing Documents" + four filter dropdowns
Main content	30% / 70%	Search Metrics (left) + Document Viewer (right)
Footer	Full width	<code>render_coresight_footer(full_width=True, stick_to_bottom=True)</code>

There are **no Streamlit tabs** on this page. Navigation is entirely via dropdowns plus the search panel.

3.2 Filter dropdowns

#	Widget key	Session key	Callback	Data source
1	<code>cf_company_select</code>	<code>cf_company</code> , <code>active_ticker</code>	<code>_on_company_change</code>	<code>_load_companies_from_db()</code> → <code>coreiq_filing_metrics_v5</code> + non-SEC blob scan
2	<code>cf_doc_type_select</code>	<code>cf_doc_type</code>	<code>_on_doc_type_change</code>	<code>_get_available_doc_types_from_db(ticker)</code>
3	<code>cf_filing_unit_select</code>	<code>cf_filing_unit</code>	<code>_on_filing_unit_change</code>	<code>get_filing_units()</code> — only for <code>FILING_UNIT_DOC_TYPES</code>
4	<code>cf_year_select</code>	<code>cf_year</code>	<code>_on_year_change</code>	<code>_get_available_years_for_doc_type(ticker, doc_type)</code>

Filing unit doc types (`app/utils/filing_units.py`): `8-K` , `6-K` , `DEF14A` , `DEFA14A` , `DEF 14A` , `PRE14A` .

Display numbers (1, 2, 3...) map to sorted `filing_N` folders; default is the **highest** filing number (most recent).

Year display: `format_func` uses `_fiscal_label_for()` to show DEI fiscal year while plumbing uses **storage/bucket year** (`COALESCE(storage_year, report_fiscal_year, fiscal_year)`).

3.3 Deep links (query params)

Param	Example	Effect
<code>ticker</code>	<code>?ticker=AAPL</code>	Resolved via <code>validate_and_get_ticker()</code>
<code>doc_type</code>	<code>?doc_type=10-Q-Q1</code>	Sets <code>cf_doc_type</code> if in <code>VALID_DOC_TYPES</code>
<code>year</code>	<code>?year=2025</code>	Sets <code>cf_year</code> if 4-digit

`st.query_params["ticker"]` is kept in sync when the user changes the company dropdown.

3.4 Left panel — Search Metrics

- Text input** — key `cf_search_input_{cf_search_gen}` ; placeholder "eg., Revenue"
- Metric results** — cards from `FilingMetricRepository.search()` ; each has **View in Document**
- Keyword hits** — `_kw_search_document()` when local file exists; separate **KEYWORD** badge section

Source badges on metric cards:

source	value	Badge
	calculated	CALC (blue)
	llm	AI (green)
	edgartools	EDGAR (orange)

3.5 Right panel — Document viewer

- Filing header** — company name, DEI fiscal year, period chip, filing date, fiscal period label
(`FilingMetricRepository.get_header_metadata()`)
- Download button** — SEC: `utils.html_to_pdf.generate_filing_pdf()` ; non-SEC: raw PDF bytes
- Viewer** — `render_sec_html_viewer()` or `_render_pdf_viewer()`

4. Session state reference

Key	Type	Purpose
<code>cf_company</code>	str	Selected ticker
<code>cf_doc_type</code>	str	DB doc type (e.g. <code>10-Q-Q1</code> , <code>annual-report</code>)
<code>cf_year</code>	str	Storage/bucket year for blob path + DB filter
<code>cf_filing_unit</code>	str	e.g. <code>filing_94</code> for multi-filing docs
<code>cf_search</code>	str	Current search term
<code>cf_search_gen</code>	int	Incremented to reset search widget on filter change
<code>cf_highlight_fact_id</code>	str None	eXtensible Business Reporting Language (XBRL) element id, <code>TEXT:...</code> , or <code>PDFPAGE:n:hit</code>
<code>cf_view_metric</code>	str None	Display label for highlight banner
<code>cf_company_select</code>	str	Widget key for company selectbox
<code>cf_doc_type_select</code>	str	Widget key for doc type
<code>cf_year_select</code>	str	Widget key for year
<code>cf_filing_unit_select</code>	int	1-based display index for filing unit
<code>active_ticker</code>	str	Global ticker used by header
<code>_highlight_just_set</code>	bool	Prevents clearing highlight on view-button rerun
<code>_view_button_click_time</code>	float	Performance timing for autoscroll
<code>_kw_highlight_keyword</code>	str	Keyword for PDF highlight overlay
<code>pdf_bytes_{ticker}_{year}_{doc_type}</code>	bytes	Generated PDF awaiting download
<code>pdf_ready_{...}</code>	bool	Triggers auto-download on next rerun
<code>pdf_b64_{path}</code>	str	Base64 PDF cached for PDF.js component
<code>_TIMING_DATA</code>	dict	Internal perf: reruns, searches, company changes

Filter change behavior: When company, doc type, year, or filing unit changes, search is cleared and highlight state reset (unless `_highlight_just_set`).

5. Components and utilities

5.1 UI components (`app/components/`)

Import	Usage
<code>styles.hide_sidebar</code>	Module load
<code>styles.render_styles</code> , <code>set_page_layout</code>	Page chrome
<code>navigation.render_header</code>	Top nav with ticker
<code>navigation.render_coresight_footer</code>	Bottom footer
<code>loading.inject_red_spinner_css</code>	Brand spinner (#D62E2F) when cache available

5.2 Domain utilities

Module	Role
<code>utils.ticker_utils</code>	<code>validate_and_get_ticker</code> , <code>DEFAULT_FALLBACK_TICKER</code>
<code>utils.filing_units</code>	Filing unit list, blob path builder, <code>FILING_UNIT_DOC_TYPES</code>
<code>utils.filings_cache</code>	DiskCache metadata (<code>get_filings_cache</code> , ETag)
<code>utils.background_scanner</code>	Background blob scan, <code>ensure_cache_purge</code>
<code>utils.non_sec_blob_fallback</code>	Parallel Azure scan for non-SEC tickers
<code>utils.html_to_pdf</code>	SEC HTML → branded PDF (PyMuPDF)
<code>utils.server_logger</code>	Structured logging (<code>log_structured_error</code> , etc.)

5.3 Data layer

Module	Role
<code>data.repository.FilingMetricRepository</code>	Metric search, header metadata, synonym expansion
<code>data.models.FilingMetricResult</code>	Search result dataclass
<code>core.llm_extractor.LLMExtractor</code>	On-demand LLM extraction
<code>core.database.db_manager</code>	Read-only queries, SQLAlchemy engine for LLM writes

5.4 Azure helpers (in-page)

`company_filings.py` implements its own Azure Blob client (not `app/utils/azure_blob.py`) with performance tuning:

- `_get_azure_config()` — env: `AZURE_STORAGE_ACCOUNT_NAME` , `AZURE_STORAGE_ACCOUNT_KEY` , `AZURE_BLOB_CONTAINER` , `AZURE_BLOB_PREFIX`
- `_get_optimized_blob_service_client()` — 64MB single-get, 8MB chunks, pool size 50
- `_ensure_local_blob_optimized()` — parallel download `max_concurrency=4` , atomic temp→rename
- `_local_cache_path_for_blob()` — maps to `data/filings_blob_cache/{relative_path}`

`app/utils/azure_blob.py` mirrors cache layout for **other pages** (e.g. earnings transcripts) and is documented here for path consistency only.

6. Database tables

6.1 `coreiq_filing_metrics_v5` (primary)

Role: Company list, year/doc_type filters, metric search, filing header metadata.

Column (conceptual)	Usage on page
<code>ticker</code> , <code>company_name</code>	Company dropdown labels
<code>doc_type</code>	Document type filter
<code>storage_year</code>	Physical Azure folder year (blob path)
<code>fiscal_year</code> , <code>report_fiscal_year</code>	DEI display labels
<code>filing_unit</code>	8-K/6-K sub-filings
<code>original_label</code> , <code>standard_concept</code> , <code>dimension_label</code>	Search targets
<code>numeric_value</code> , <code>unit_ref</code> , <code>value</code>	Display + formatting
<code>ixbri_id</code>	HTML autoscroll target (<code>document.getElementById</code>)
<code>source</code>	<code>xbri</code> , <code>calculated</code> , <code>llm</code> , <code>edgartools</code> , <code>store_count</code> , <code>credit_rating</code>
<code>llm_query</code>	Formula string (calculated) or JSON detail (store/credit)
<code>period_*</code> , <code>statement_type</code>	Card metadata
<code>filing_date</code> , <code>filing_date_sec</code>	Header metadata

Indexes referenced in code: `idx_ticker_fiscal_doctype` (covering prefetch).

6.2 `coreiq_filing_metrics_v2`

Used by:

- `filing_units.get_filing_units_from_db()` — `DISTINCT filing_unit` for 8-K/6-K
- `llm_extractor.insert_llm_metric()` — inserts new LLM rows with `source='llm'`

Runtime search reads **v5**; LLM writes go to **v2** (legacy split).

6.3 `filing_llm_cache`

Column	Purpose
<code>ticker</code> , <code>fiscal_year</code> , <code>doc_type</code> , <code>query_normalized</code>	Cache key
<code>status</code>	<code>found</code> <code>not_found</code>
<code>metric_id</code>	Link to inserted v2 row
<code>searched_at</code>	Timestamp

6.4 coreiq_companies

Non-SEC company names via `get_non_sec_tickers_from_db()` in `utils.non_sec_blob_fallback`.

6.5 coreiq_ir_websites / company_ir_websites

Not used directly on this page. IR URLs are resolved in `EarningsCalendarRepository._build_ir_website_lookup()` (`app/data/repository.py`) for the earnings calendar. Documented here because filing metrics pipeline and IR discovery share the same company universe.

7. Document types and Azure blob layout

7.1 VALID_DOC_TYPES

SEC: `10-K`, `10-Q-Q1` ... `10-Q-Q3`, `8-K`, `20-F`, `6-K`, `DEF14A`, `DEFA14A`, `DEF 14A`, `PRE14A`

Non-SEC (Capital Q): `annual-report`, `interim-report-Q1` ... `Q5`, `half-yearly`

7.2 Blob path conventions

```
{AZURE_BLOB_PREFIX}/{TICKER}/{YEAR}/{DOC_TYPE}/filing.html      # SEC default
{AZURE_BLOB_PREFIX}/{TICKER}/{YEAR}/{DOC_TYPE}/filing_94/filing.html  # 8-K unit
{AZURE_BLOB_PREFIX}/{TICKER}/{YEAR}/{annual-report}/filing.pdf  # non-SEC
```

Parser: `_parse_blob_path()` expects at least 4 path segments after prefix.

Scanner: `scan_filings_directory()` groups blobs by `(ticker, year, doc_type_dir)` and picks best file via `_pick_best_blob_for_docdir()` (prefers `filing.html` / `filing.pdf`).

7.3 SEC vs non-SEC routing

Routing is by **doc_type**, not company flag:

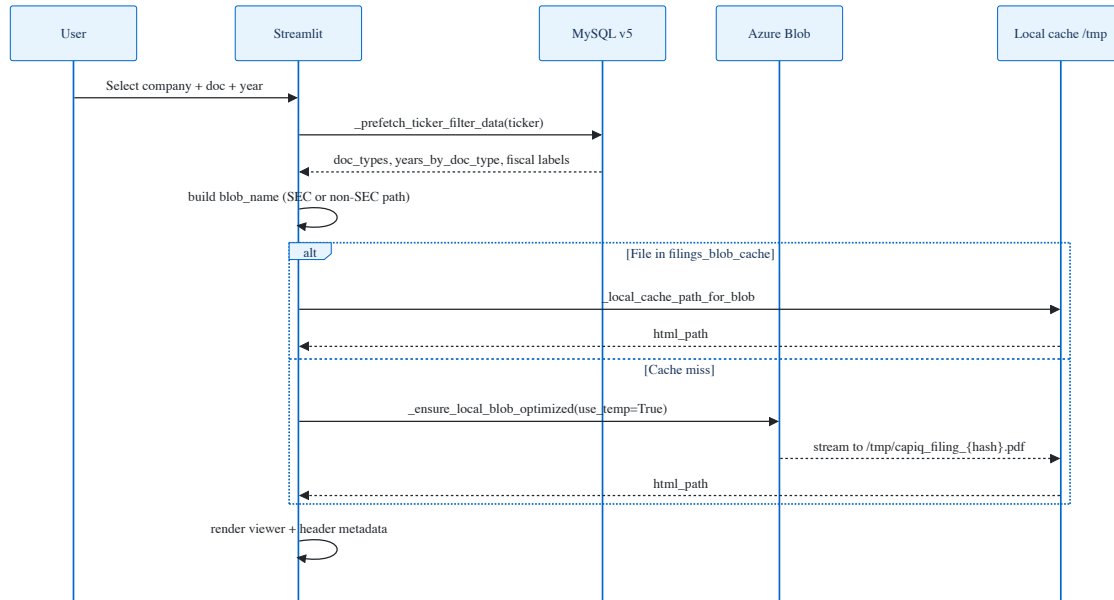
Doc type set	File	Viewer
<code>NON_SEC_DOC_TYPES</code>	<code>filing.pdf</code>	<code>_render_pdf_viewer</code> (PDF.js)
All other valid SEC types	<code>filing.html</code>	<code>render_sec_html_viewer</code>

10-Q storage year fallback: If blob missing at `{year}`, retries `{year-1}` for non-December fiscal year ends (e.g. LULU).

8. Filing load flow

Selecting a filing loads metrics from XBRL tables, with optional LLM extraction on cache miss.

Selecting a filing loads metrics from XBRL tables, with optional LLM extraction on cache miss.



9. XBRL / iXBRL handling

9.1 Ingestion (offline)

XBRL facts are extracted during ETL into `coreiq_filing_metrics_v5` with `source='xbrl'`. Calculated metrics use `source='calculated'`. Some supplemental metrics use `edgartools` during batch jobs (see `app/data/repository.py` segment/ratings paths) — not invoked from this page.

9.2 iXBRL HTML processing (runtime)

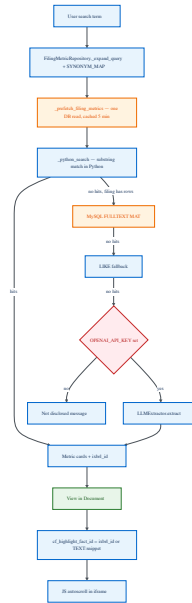
`_load_and_process_html()` pipeline:

1. Read UTF-8 from local path
2. Strip XML declaration (force HTML5 mode)
3. Remove `<script>` tags (malformed SEC JS breaks iframe)
4. `_convert_ixbrl_to_spans()` — namespace tags → `<span>` with preserved `id` for `ixbrl_id` navigation
5. Inject `content-visibility: auto` CSS for lazy rendering
6. `_resolve_sec_edgar_base_url()` — rewrite relative links to SEC EDGAR absolute URLs
7. Inject click handler: hash links scroll in-iframe; external links open new tab

9.3 XBRL extract / metric search flow

Metric search checks filing tables first, then may invoke GPT-4o-mini extraction for missing values.

Metric search checks filing tables first, then may invoke GPT-4o-mini extraction for missing values.



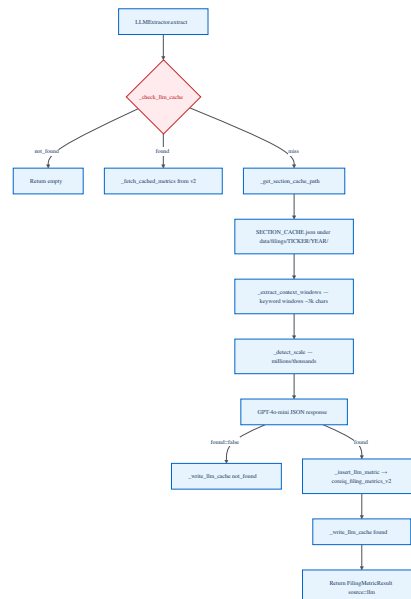
9.4 FilingMetricRepository.search() strategy

1. Expand query with **SYNONYM_MAP** (100+ financial terms)
2. **Fast path:** prefetch all numeric rows for (ticker, year, doc_type) → Python filter/dedup/sort (0ms after first call)
3. **DB fallback:** `MATCH ... AGAINST ON (original_label, standard_concept, dimension_label) then LIKE`
4. Dedup: latest period_end / period_instant per (label, dimension, dimension_label)
5. Exclude standard_concept LIKE '%Text Block'

10. LLM extraction (app/core/llm_extractor.py)

Triggered from `main()` when `FilingMetricRepository.search()` returns empty and `OPENAI_API_KEY` is set.

Triggered from `main()` when `FilingMetricRepository.search()` returns empty and `OPENAI_APIKEY` is set.



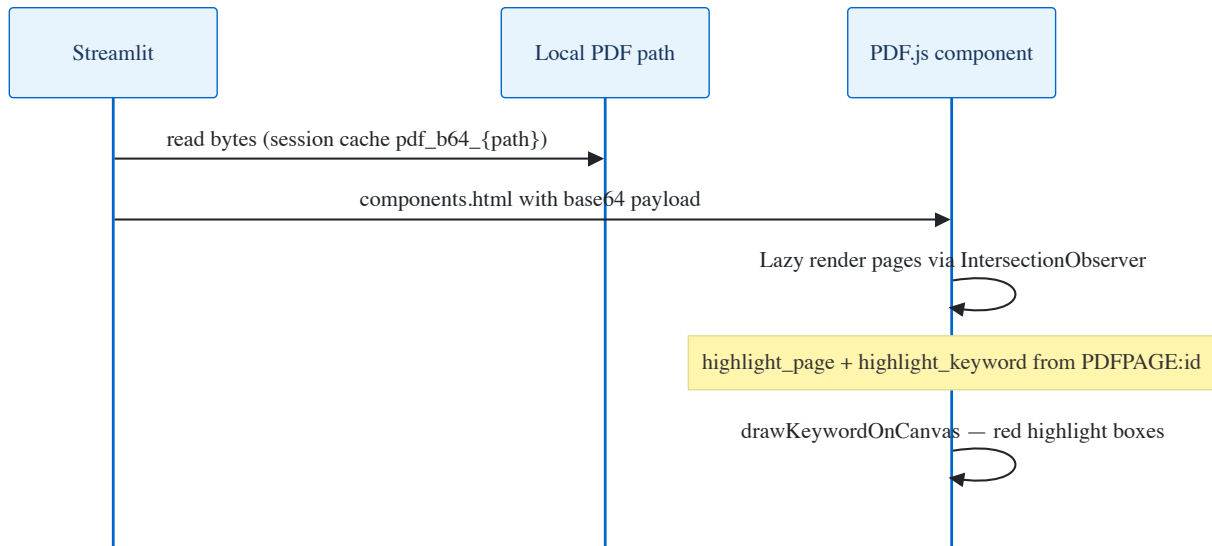
Cost note (from module docstring): ~\$0.00012 per unique query; cached queries \$0.

SECTION_CACHE.json sections prioritized: `part_ii_item_8`, `part_ii_item_7`, `part_i_item_1`, etc.

11. PDF viewing flow (non-SEC)

Non-SEC PDFs stream from Azure Blob via SAS URL embedded in the Streamlit viewer.

Non-SEC PDFs stream from Azure Blob via SAS URL embedded in the Streamlit viewer.



Why PDF.js: Chrome blocks native PDF plugins inside Streamlit's sandboxed iframe. PDF.js renders to `<canvas>` without plugins.

Text index: `_build_doc_text_index()` uses PyMuPDF (`fitz`) per page; `_kw_search_document()` returns snippets with physical + printed page labels.

12. SEC HTML → PDF download flow

SEC HTML filings are converted server-side to PDF for download via the PDF utility.

SEC HTML filings are converted server-side to PDF for download via the PDF utility.



Non-SEC downloads skip generation — existing `filing.pdf` bytes are served directly.

13. Caching and performance

Layer	Mechanism	TTL / scope
Streamlit <code>@st.cache_data</code>	Companies, prefetch, HTML process, text index, keyword search	300–3600s
Streamlit <code>@st.cache_resource</code>	Azure clients, <code>scan_filings_directory</code>	Process lifetime
<code>filings_cache</code> DiskCache	Per-ticker blob metadata from background scanner	Disk-backed
<code>data/filings_blob_cache/</code>	Downloaded blobs (when not <code>use_temp</code>)	Filesystem + mtime vs blob <code>last_modified</code>
<code>/tmp/capiq_filing_*</code>	On-demand downloads (<code>use_temp=True</code>)	OS-managed
Session state	PDF base64, generated PDF bytes	Per session

Load order in `main()` : Header renders first → companies load → session init → dropdowns → two-column content. Early spinner: "Loading filing data..."

14. Authentication and access

- Identity via `main.py` OIDC cookie hydrate; optional `require_auth(page="company_filings")` at page level.
- Page is registered without ACL flag `False` in `main.py` — default authenticated access.

15. Error handling

- Module-level `try/except` around `main()` shows generic Streamlit error
- Azure download failures: warning + retry button
- Missing filing: `st.error` with blob path + placeholder viewer
- All significant exceptions logged via `log_structured_error` with `page="company_filings"`

16. Environment variables

Variable	Purpose
<code>AZURE_STORAGE_ACCOUNT_NAME</code>	Blob account (default <code>csmarketdata</code> in code)
<code>AZURE_STORAGE_ACCOUNT_KEY</code>	Blob credential
<code>AZURE_BLOB_CONTAINER</code>	Container name (default <code>azure-storage-test</code>)
<code>AZURE_BLOB_PREFIX</code>	Optional path prefix inside container
<code>OPENAI_API_KEY</code>	Enables LLM metric fallback

17. Related files (quick index)

Path	Relationship
<code>app/pages/company_filings_add_files.py</code>	Admin Azure upload for non-SEC PDFs
<code>app/utils/azure_blob.py</code>	Shared blob cache helpers (transcripts)
<code>app/utils/filing_units.py</code>	Multi-filing path logic
<code>app/data/repository.py</code>	<code>FilingMetricRepository</code> class
<code>app/core/llm_extractor.py</code>	LLM on-demand extraction
<code>app/utils/html_to_pdf.py</code>	SEC download PDF generation
<code>edgartools/</code> (repo root)	Offline SEC parsing; not imported by this page

18. Key functions index (`company_filings.py`)

Function	Responsibility
<code>main()</code>	Page entry, session init, layout
<code>_on_company_change</code> / <code>_on_doc_type_change</code> / <code>_on_year_change</code> / <code>_on_filing_unit_change</code>	Filter callbacks
<code>_prefetch_ticker_filter_data</code>	Single-query filter data per ticker
<code>_get_non_sec_blob_path</code>	Non-SEC PDF blob path
<code>build_filing_blob_path</code> (filing_units)	SEC path with filing unit
<code>_ensure_local_blob_optimized</code>	Azure download
<code>FilingMetricRepository.search</code>	Metric search
<code>render_sec_html_viewer</code>	SEC HTML iframe + autoscroll
<code>_render_pdf_viewer</code>	PDF.js viewer
<code>_kw_search_document</code>	Keyword hits in document
<code>scan_filings_directory</code>	Full Azure inventory (fallback company list)